

Contour in Image Processing

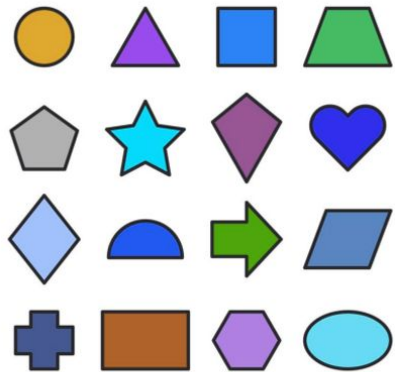
- Curves joining all continuous points with the same intensity.
- Represent object boundaries
- Difference from edges: edges show local intensity changes; contours trace shape with continuous outlines.

Extraction Steps:

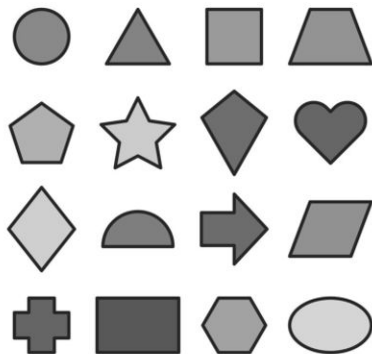
- Convert to grayscale.
- Denoise (Gaussian blur).
- Apply thresholding or Canny edge detection.
- Use `cv2.findContours()` in OpenCV

Example

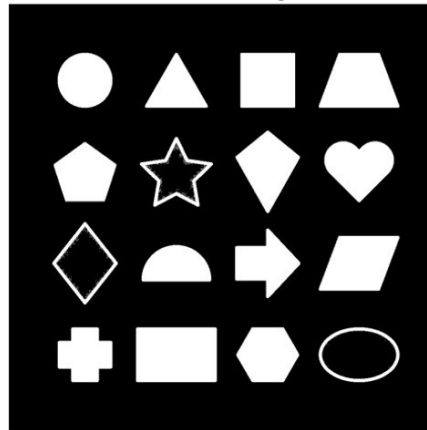
Original Image



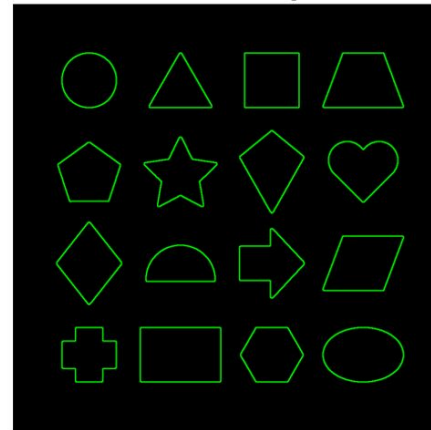
Grayscale Image



Thresholded Image



Contours on Black Background



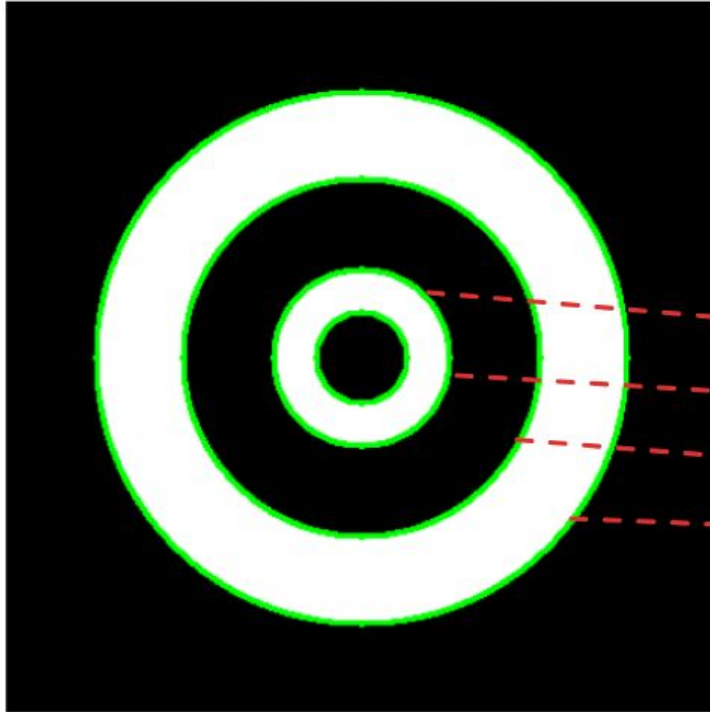
cv.findContours() function

cv.findContours() function parameters,

- first one is source image,
- second is contour retrieval mode(e.g. cv2.RETR_LIST, cv2.RETR_CCOMP, RETR_TREE), RETR_LIST returns all contours in list format, RETR_CCOMP returns in two-level hierarchical format, RETR_TREE
- third is contour approximation method(e.g. cv2.CHAIN_APPROX_SIMPLE), Which it used to save memory by compressing points.

Without hierarchical format contouring

Contours using cv2.RETR_LIST



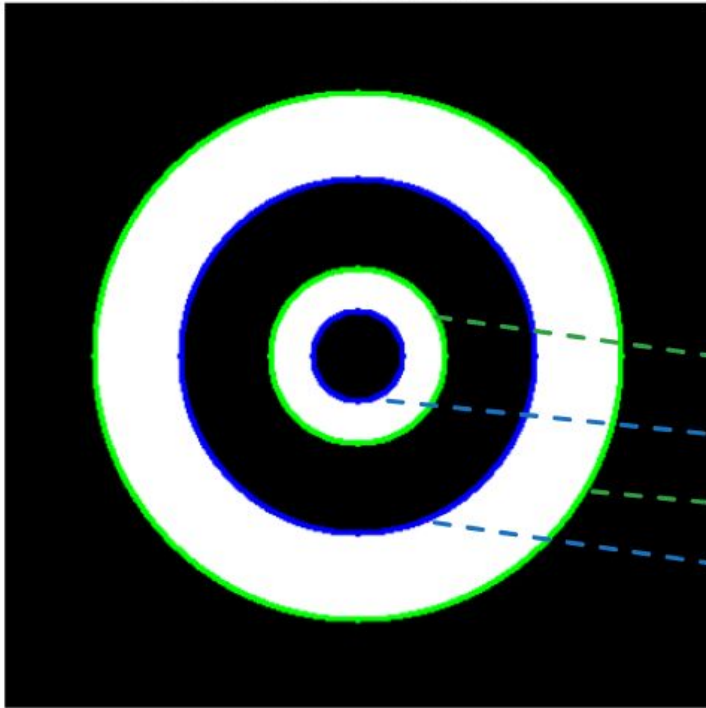
Hierarchy

[Next Previous First_Child Parent]

→	[[	1	-1	-1	-1	]	→	Level 0, Contour 0
→	[	2	0	-1	-1	]	→	Level 0, Contour 1
→	[	3	1	-1	-1	]	→	Level 0, Contour 2
→	[	-1	2	-1	-1	]]]	→	Level 0, Contour 3

2 level hierarchical format contouring

Contours using cv2.RETR_CCOMP



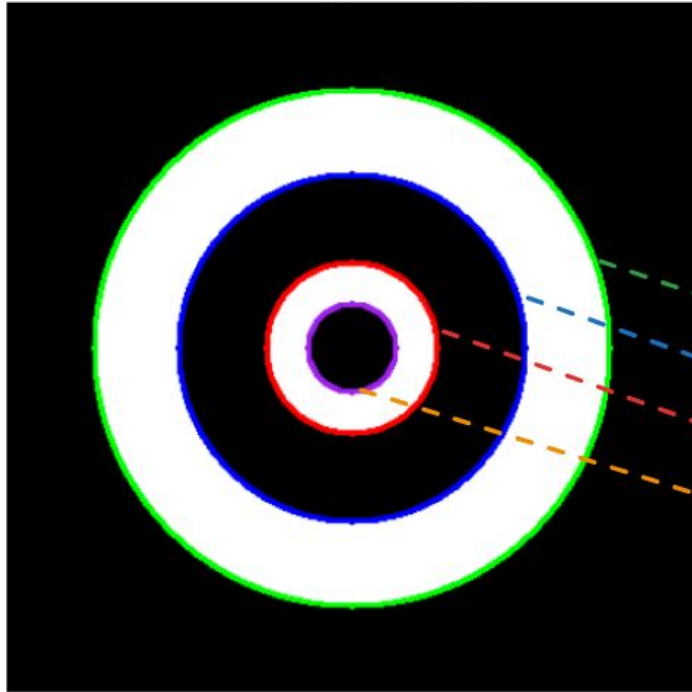
Hierarchy

[Next Previous First_Child Parent]

→	[[[	2	-1	1	-1	]	→	Level 0, Contour 0
→	[	-1	-1	-1	0	]	→	Level 1, Contour 1
→	[	-1	0	3	-1	]	→	Level 0, Contour 2
→	[	-1	-1	-1	2	]]]	→	Level 1, Contour 3

Tree hierarchical format contouring

Contours using cv2.RETR_TREE



Hierarchy

[Next Previous First_Child Parent]

→	[[[	-1	-1	1	-1	]	→	Level 0, Contour 0
→	[	-1	-1	2	0	]	→	Level 1, Contour 1
→	[	-1	-1	3	1	]	→	Level 2, Contour 2
→	[	-1	-1	-1	2	]]]	→	Level 3, Contour 3